BUSHRA BUSINESS BANK
NETWORK ANALYSIS AND DESIGN

Proposed Enterprise Network Design Report

GROUP MEMBERS

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Scope: Planning and documentation only — no configuration

1. Executive Summary

This report proposes a secure, scalable and manageable network for Bushra Business Bank. The design uses a hierarchical star topology with a redundant core, separated department VLANs, a protected server segment, an isolated DMZ, secure site-to-site VPN connectivity for two assumed branches, and redundant Internet connectivity. The design is conceptual and contains no device configuration, command-line instructions or implementation.

2. Project Assumptions and Scope

Planning assumptions: one head office in Mogadishu and two smaller branches; approximately 150–200 head-office endpoints initially; 30–50 endpoints per branch; central banking applications hosted in a protected server segment; and allowance for future growth. Replace these assumptions if actual figures are provided.

In scope: requirements analysis, topology proposal, device roles, VLAN and IP plan, security architecture, resilience and explanation. Out of scope: router/switch/firewall configuration, CLI commands, cabling installation and production deployment.

3. Company Overview

Bushra Business Bank is treated in this academic scenario as a financial institution with a head office and two branches. Its network supports staff communication, customer service, finance operations, secure access to banking applications, branch connectivity, printing, monitoring and controlled guest Internet access.

4. Network Problem Statement

• Department traffic may be mixed on a flat network, increasing broadcast traffic and complicating access control.

• Sensitive Finance, HR and banking systems require separation from general staff and guest devices.

• Branch users need dependable access to central services without exposing internal systems to the public Internet.

• A single network device or Internet connection could create an avoidable outage.

• Without centralized monitoring, backups and documented IP allocation, troubleshooting and growth become harder.

5. Proposed Solution and Design Goals

• Hierarchical edge, core and access layers.

• Department-based VLANs and subnet boundaries.

• DMZ for approved public-facing services.

• Next-generation firewalls, access-control policies and encrypted VPNs.

• Redundant core switches and dual ISP links where budget permits.

• Central identity, DNS, DHCP, logging, monitoring and backup services.

6. Network Topology

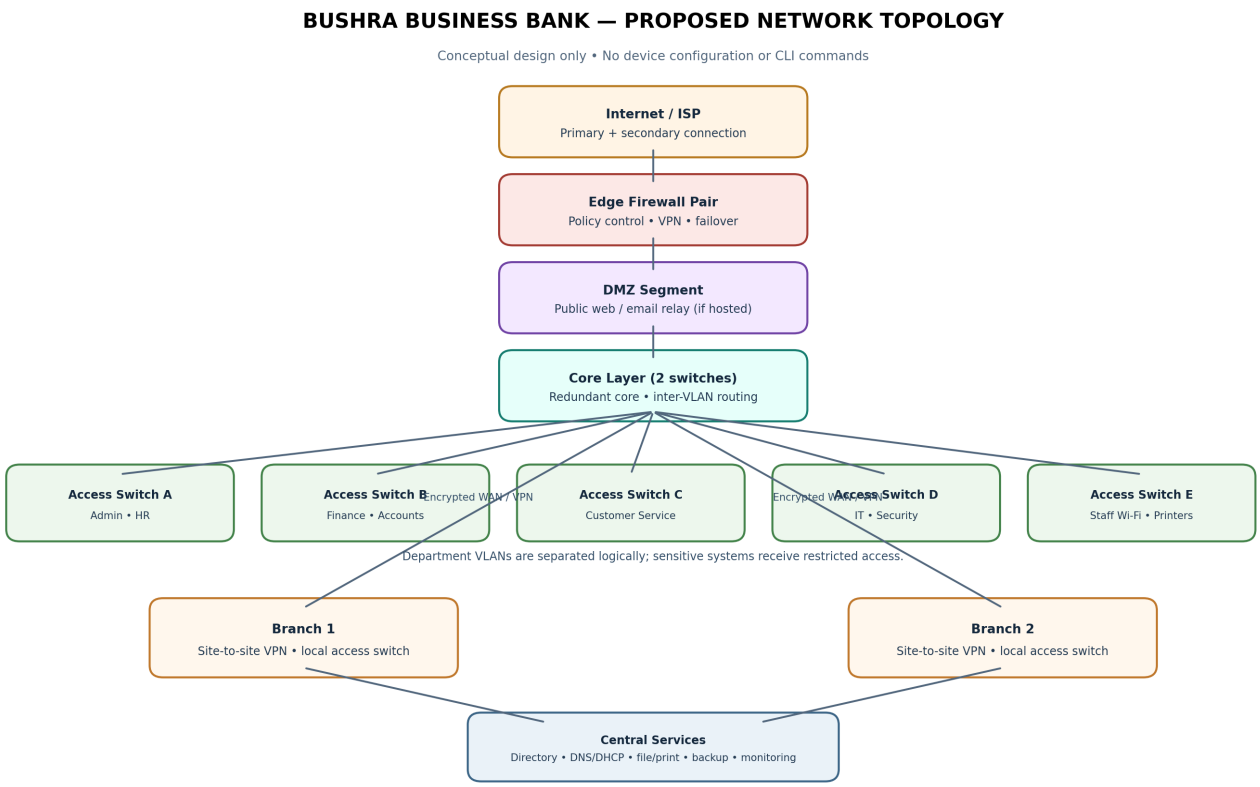


Figure 1. Conceptual topology for the head office and two branches. Department VLANs are logically separated; branches connect through encrypted site-to-site VPN tunnels.

7. Network Devices and Their Roles

Device	Purpose
2 edge firewalls (HA pair)	Perimeter security, VPN termination, policy enforcement and failover.
2 core Layer-3 switches	High-speed backbone, inter-VLAN routing and resilient core connectivity.
Managed access switches	Connect PCs, printers, access points, cameras and phones; support VLANs.
Wireless access points	Provide managed staff Wi-Fi and isolated guest Wi-Fi.
2 ISP connections	Primary and backup Internet paths to improve availability.
Servers / virtual host	Directory and identity, DNS, DHCP, file services, application services, logging and backup.
UPS and rack equipment	Power protection and organized physical installation.
Monitoring / security tools	Central log collection, network monitoring and alerting.

8. VLAN and IP Addressing Plan

Private IPv4 space 10.10.0.0/16 is reserved for the bank. The first usable address in each subnet is the proposed default gateway. Usable host counts exclude network and broadcast addresses.

VLAN / Site	Department / Purpose	Network / Prefix	Gateway	Usable hosts	Notes
VLAN 10	Administration	10.10.10.0/26	10.10.10.1	62	Staff PCs and admin printers
VLAN 20	Finance & Accounts	10.10.20.0/26	10.10.20.1	62	Restricted financial workstations
VLAN 30	Customer Service	10.10.30.0/25	10.10.30.1	126	Teller/customer-service endpoints
VLAN 40	IT Department	10.10.40.0/27	10.10.40.1	30	IT support and management
VLAN 50	Human Resources	10.10.50.0/27	10.10.50.1	30	HR workstations
VLAN 60	Security & CCTV	10.10.60.0/26	10.10.60.1	62	Cameras and physical-security systems
VLAN 70	Staff Wi-Fi	10.10.70.0/25	10.10.70.1	126	Managed staff wireless devices
VLAN 80	Guest Wi-Fi	10.10.80.0/24	10.10.80.1	254	Internet-only guest access
VLAN 90	Servers / Data Centre	10.10.90.0/27	10.10.90.1	30	Directory, DNS, DHCP, file and app servers
VLAN 100	Printers / IoT	10.10.100.0/27	10.10.100.1	30	Network printers and approved IoT
VLAN 110	DMZ	10.10.110.0/28	10.10.110.1	14	Public-facing services, isolated from LAN
VLAN 120	Network Management	10.10.120.0/27	10.10.120.1	30	Management interfaces; IT-only

VLAN / Site	Department / Purpose	Network / Prefix	Gateway	Usable hosts	Notes
Branch 1	Branch 1 users	10.10.130.0/24	10.10.130.1	254	Branch user network; VLAN split locally if required
Branch 2	Branch 2 users	10.10.140.0/24	10.10.140.1	254	Branch user network; VLAN split locally if required
WAN links	Point-to-point links	10.10.250.0/30 blocks	Per-link	2 usable each	Allocate a separate /30 per routed point-to-point link

9. Subnetting Explanation

A /26 has 64 total addresses and 62 usable; /25 has 128 total and 126 usable; /27 has 32 total and 30 usable; /28 has 16 total and 14 usable; /24 has 256 total and 254 usable. Larger subnets are assigned to teams with more devices.

Example — VLAN 20 Finance: Network 10.10.20.0/26; proposed gateway 10.10.20.1; usable host range 10.10.20.1–10.10.20.62; broadcast 10.10.20.63. Reserve .1 for the gateway and document DHCP/static allocations.

10. Security Design

• Separate Finance, HR, IT, CCTV, guest Wi-Fi, servers and management traffic into distinct VLANs.

• Apply least-privilege access policies.

• Guest Wi-Fi reaches the Internet only and cannot access internal bank networks.

• Use MFA for remote and privileged access.

• Use encrypted site-to-site VPNs between branches and head office.

• Place public-facing systems in a DMZ and restrict DMZ-to-LAN communication.

• Use endpoint protection, patching, secure backups and centralized logging.

• Restrict network-device management to authorized IT workstations and the management VLAN.

11. Availability, Performance and Scalability

Redundant core switches, a high-availability firewall pair and primary/secondary ISP links reduce single points of failure. Managed switches and structured cabling support expansion. QoS may prioritize approved business applications. UPS protection and tested backups support continuity. Branches can expand while retaining the reserved 10.10.0.0/16 plan.

12. Implementation Considerations — Design Only

Before deployment, confirm actual users, applications, branches, bandwidth, equipment, regulatory obligations and budget. A site survey and security review should validate the design. Qualified personnel should implement and test it under an approved change plan. This assignment stops at design and includes no configuration.

13. Conclusion

The proposed network provides a structured foundation for secure communication between departments and branches. VLAN separation, subnet planning, perimeter protection, a DMZ, encrypted VPNs and redundancy address the assumed network problems while allowing future growth.

14. References

• RFC 1918 — Address Allocation for Private Internets.

• RFC 4632 — Classless Inter-domain Routing (CIDR).

• Cisco Networking Academy — Networking Basics and network design learning materials.

• Course notes: Network Analysis and Design.

Appendix A. Group Submission Checklist

■ Report reviewed by all five members

■ Assumptions confirmed with lecturer

■ Topology diagram included

■ VLAN and IP plan checked for overlaps

■ No configuration commands included

■ Files placed in one folder and ZIP archive created

■ GitHub repository link checked